

CAPSTONE PROJECT

PHASE-1

DOCUMENTATION

Online Education Drowsiness Detection System Using Machine Learning

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ABSTRACT

Drowsiness is a critical issue impacting safety, productivity, and engagement in various domains, including driving, online learning, and workplaces. This project presents a real-time drowsiness detection and engagement monitoring system using advanced machine learning techniques. The system analyses facial and behavioural cues like eye closure, yawning, and attention span, providing real-time alerts and detailed reports. Leveraging highly accurate supervised learning models, such as K-Nearest Neighbours (KNN) with 97.5% precision and Support Vector Machines (SVM), the solution enhances focus, prevents fatigue-related risks, and boosts efficiency. Designed for multi-scenario adaptability, it operates with minimal hardware, utilizing standard webcams and Python-based frameworks like OpenCV and Scikit-learn. The system's modular approach includes data acquisition, feature extraction, classification, real-time alerts, and engagement analytics, making it a scalable and effective tool for fatigue management.

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